SeungHun Lee

+82 (0)10-7495-7445 • Mr.leesh90@gmail.com • https://leesh90.github.io/about/

Professional Experience

3billion, Seoul, South Korea

AI Engineer March 2026 - Present

- Developing foundation models for DNA, RNA, and protein sequences.
- Applying deep learning architectures for large-scale biological sequence modeling.
- Building scalable training pipelines for large-scale genomic datasets.

Current Research Interests

My research focuses on AI-driven modeling of biological sequences, particularly in genomics and molecular biology. I develop foundation models to understand genomic function and predict the impact of genetic variants. I am particularly interested in large-scale sequence modeling and representation learning for DNA/RNA/protein data, along with explainable AI (XAI) techniques to improve model interpretability in high-stakes domains.

Education

- **Sangmyung University, Cheonan, South Korea (Advisor: Prof. Hyun-chul Kim)**
- Ph.D., Computer Software Engineering, GPA: 4.45/4.5 March 2018 – Aug 2026
- M.S., Computer Software Engineering, GPA: 3.68/4.5 March 2015 - Feb 2017
- B.S., Computer Software Engineering, GPA: 4.24/4.5 March 2009 - Feb 2015

Doctoral Research

My doctoral research spanned digital finance, network security, and Explainable AI (XAI). In digital finance, I employ a multimodal approach (i.e., text, video, and audio) and user behavior data to analyze crowdfunding platforms, detecting fraudulent campaigns and identifying success factors. For network traffic, I focus on traffic classification and intrusion detection, analyzing both traffic headers and payloads while developing methods to classify traffic patterns. Across these domains, I leverage advanced AI methodologies including natural language processing, pattern analysis, classical machine learning algorithms, transformers, graph neural networks (GNNs), large language models (LLMs), and Agent AI systems. Throughout my work, I integrate XAI techniques to enhance the reliability, transparency, and interpretability of AI-driven solutions.

Publications

1. Seunghun Lee, Muneeb A. Khan, and Hyun-Chul Kim. “Explainable Prediction of Crowdfunding Success Using Hierarchical Attention Network.” MDPI electronics, vol. 15, no. 3, Jan. 2026.

2. Seunghun Lee, Heemin Park, and Hyun-Chul Kim. "Fraud Detection on Crowdfunding Platforms Using Multiple Feature Selection Methods." IEEE Access, Mar. 2025.

3. Seunghun Lee, Wafa Shafqat, and Hyun-chul Kim. "Backers beware: Characteristics and detection of fraudulent crowdfunding campaigns." Sensors, vol. 22, no. 19, Oct. 2022.

4. Kang-Hee Lee, Seunghun Lee, and Hyun-Chul Kim. "Traffic classification using deep learning: being highly accurate is not enough." In Proceedings of the SIGCOMM, Aug. 2020. (poster)

5. Seunghun Lee, KangHee Lee, and Hyun-chul Kim. "Content-based success prediction of crowdfunding campaigns: A deep learning approach." In Proceedings of the ACM conference on Computer Supported Cooperative Work and Social Computing (CSCW), Nov. 2018. (poster)

6. Seunghun Lee, and Hyun-chul Kim. "Crowdfunding Scams: The Profiles and Language of Deceivers." Journal of The Korea Society of Computer and Information, vol. 23, no.3, Mar. 2018.

7. Kang-hee Lee, Seunghun Lee, and Hyun-chul Kim. "Predicting Success of Crowdfunding Campaigns using Multimedia and Linguistic Features." Journal of Korea Multimedia Society, vol. 21, no. 2, Feb. 2018.

8. Wafa Shafqat, Seunghun Lee, Sehrish Malik, and Hyun-chul Kim. "The language of deceivers: Linguistic features of crowdfunding scams." In Proceedings of the International Conference Companion on World Wide Web(WWW), Apr. 2016 (poster)

9. Yoon, Jangho, Seunghun Lee, and Hyun-chul Kim. "Smart SNS Map: Location-based Social Network Service Data Mapping and Visualization System." Journal of Korea Multimedia Society, vol. 19, no. 2, Feb. 2016.

10. Seunghun Lee, Daeyoung Oh, Minhyuk Kang, and Hyun-chul kim. "SNS Map: Location-based SNS data mapping system." In Proceedings of the Korea Computer Congress (KCC), June. 2015.

Projects

Explainability in Graph Neural Networks for Internet Traffic Classification June 2023 – Feb 2025
- Trained and classified encrypted traffic using GNNs on host behavior and network-wide graphs.
- Applied graph XAI methods (e.g., GNNExplainer) to reveal distinctive graph structural patterns.

Research on Virtualization-based 5G Networks and Cyber Threats April 2023 – Oct 2023
- Developed Cyber5Gym, a virtualization-based multi-agent training environment for 5G networks using Open5GS and UERANSIM.
- Implemented cyber threat scenarios including high-bandwidth traffic, SMC replay, and DDoS attacks.

Deep Learning based Internet Traffic Classification: Myths, Realities, and their Explainabilities
June 2022 – May 2023
- Applied Hierarchical Attention Networks (HAN) to Internet traffic classification.
- Used attention visualization to explain and identify key traffic features.

Research on Building a Virtualization-Based 5G Cyber Training Environment April 2022 – Oct 2022
- Built a virtualization-based 5G cyber training environment using Open5GS and UERANSIM.
- Investigated major 5G vulnerabilities, detailing attack procedures, impacts, and countermeasures.

Multi-modal data-driven Explainable AI Systems and the Future of Digital Finance Sep 2019 – Feb 2022
- Analyzed text using HAN with attention visualization for feature interpretation.
- Applied CNNs and saliency maps for video and image analysis.
- Extracted acoustic features via MFCC and modeled metadata with MLP.
- Integrated all modalities into a unified multi-modal XAI system for fraud detection in digital finance.

Towards Explainable AI in Next-Generation Intrusion Detection Systems April 2019 – Oct 2019
- Applied HAN to network traffic classification using its hierarchical flow–packet–byte structure.
- Used attention visualization to identify salient header fields and payload patterns.

Statistics-based Network Behavior Modeling May 2018 – Oct 2018
- Modeled network behavior using host analysis and Latent Dirichlet Allocation (LDA) for anomaly detection.
- Analyzed and visualized anomalous patterns via host behavior graphs and radar charts.

Traffic Measurement in Anonymity Networks April 2017 – Oct 2017
- Collected and analyzed network traffic from the Tor anonymity network.

Characterization and Automatic Labeling of Malicious Traffic in Control System Networks
April 2017 – Oct 2017
- Applied Latent Dirichlet Allocation (LDA) to classify Industrial Control System (ICS) network traffic.
- Modeled traffic flows as documents to automatically extract signatures and label unclassified SCADA traffic.

Network Traffic Classification for Intrusion Detection June 2015 – Dec 2015

- Applied LDA for automated signature detection and classification of general application network traffic.

Honors and Awards

Teaching Assistant Scholarship, Computer Software Engineering, Sangmyung Univ.	2019 - 2020
Research Assistant Scholarship, Computer Software Engineering, Sangmyung Univ.	2015 – 2016
Second Honors Graduate, B.S., Computer Software Engineering, GPA: 4.24/4.5	2015
Award for Creative Design Competition (Graduation Project), College of Engineering	2014

Teaching Experience

Introduction to Data Science (Big Data), Teaching Assistant and Lab Instructor	2019 – 2025
Algorithms, Teaching Assistant	2018 – 2024
Computer Programming, Teaching Assistant and Lab Instructor	2017, 2019